

QUESTION_37

Aim

To analyze the time complexity of the given recurrence relation using the Master Theorem, identify the appropriate case, and evaluate the chatbot's response time as the number of user queries increases.

Procedure

1. Consider the given recurrence relation.
2. Identify the values of a , b , and $f(n)$.
3. Compare $f(n)$ with the standard Master Theorem term.
4. Determine the appropriate Master Theorem case.
5. Find the asymptotic time complexity.
6. Analyze the chatbot's performance as the number of user queries increases.
7. Conclude the efficiency of the AI-based chatbot.

Algorithm

Step 1: Start.

Step 2: Consider the recurrence relation:

- $T(n) = T(n/2) + \log n$

Step 3: Identify the parameters:

- $a = 1$
- $b = 2$
- $f(n) = \log n$

Step 4: Compare $f(n)$ with the standard Master Theorem term.

Step 5: Since $f(n)$ grows as $\log n$, it satisfies Master Theorem Case 2.

Step 6: Determine the time complexity as:

- $\Theta((\log n)^2)$

Step 7: Analyze the chatbot's response time for increasing numbers of user queries.

Step 8: Stop.

Input

Recurrence Relation:

- $T(n) = T(n/2) + \log n$

Output

- $a = 1$
- $b = 2$
- $f(n) = \log n$
- Master Theorem Case: Case 2
- Time Complexity: $\Theta((\log n)^2)$

Result

The given recurrence relation satisfies Master Theorem Case 2. Therefore, the overall time complexity of the AI-based chatbot processing algorithm is $\Theta((\log n)^2)$.